

Isaac Lee

Anaheim, CA | (714) 715 3276 | iscl1706@gmail.com

Education

University of California, Irvine – BS, Mechanical Engineering

Expected June 2028

GPA: 3.56 | Honors: Dean's Honor List

- **Relevant Coursework:** Fluid Dynamics, Mechanics of Structures, Thermodynamics, Statics, Dynamics

Experience

Engineering Intern, Sierra Aluminum – Fontana, CA

Sept 2026 – Present

- Reduced paint-line scrap from 10% to 4% by analyzing aluminum extrusion manufacturing processes, production data, and defect trends to identify sources of waste
- Ensured aluminum extrusions met dimensional and manufacturing requirements by reviewing engineering drawings, technical specifications, and inspection data
- Worked directly on the production floor with engineering and manufacturing teams to investigate process issues, inspect parts, and support production improvements
- Supported material testing and process validation through tensile testing, equipment calibration, and analysis of test data in Excel

Projects

Aerodynamics Engineer, FSAE Electric Racing

Apr 2026 – Present

- Designed and optimized a carbon fiber undertray in SolidWorks to achieve a 40 N downforce target, refining throat and diffuser geometry, underbody sealing, and integration with sidepod and bodywork
- Analyzed pressure coefficient, streamlines, crossflow, and flow separation using STAR-CCM+ CFD to guide design and geometry choices and iterations and reduce separation or choking through the diffuser and throat
- Manufactured carbon fiber components through wet layup and vacuum bagging, controlling fiber orientation, resin saturation, and vacuum pressure to improve layup consistency and minimize manufacturing defects

Hopper Robot

Apr 2026 – Jun 2026

- Led mechanical design and fabrication of an autonomous pneumatic-propulsion robot within a 3-person team, delivering a fully integrated mechanical system for final testing
- Manufactured structural and mounting components using woodworking, laser cutting, and additive manufacturing, applying GD&T and tolerancing principles to ensure proper component fit and assembly
- Iterated chassis and mounting designs through rapid prototyping and testing to improve structural rigidity, component integration, and resistance to tipping

Autonomous Rover

Sept 2024 – Mar 2025

- Fabricated polycarbonate and FDM 3D-printed chassis components using a jigsaw, bandsaw, and drill press, creating a rigid platform for drivetrain, electronics, and sensor integration
- Designed and assembled the rover electrical system by developing wiring diagrams, soldering components, and integrating motors, sensors, servos, and Arduino controls
- Tuned sensor response and rover behavior through iterative testing to reduce line-following overcorrection and verify drivetrain, electronics, and control-system functionality before final trials

Skills

CAD/Design: SolidWorks (CSWA Certification), OnShape, AutoCAD, GD&T, Engineering Drawings

Analysis/Simulation: Ansys Mechanical (FEA), Star-CCM+ (CFD), SolidWorks Simulation

Manufacturing: Composite Manufacturing, CNC Machining, Laser Cutting, FDM 3D Printing, Soldering

Programming/Software: MATLAB, C++, Java, JavaScript

Documentation: Microsoft Office (Excel, Word, PowerPoint), Technical Writing